

The cross entropy between two probability distributions p and q is defined as

$$\mathbb{H}_{ce}(p, q) = -\mathbb{E}_{X \sim p} [\log q(X)] = -\sum_x p(x) \log q(x)$$

This measures the average number of bits needed to encode events from p using a code optimised for q . The cross entropy is always greater than or equal to the entropy of p , with equality when $p = q$, in which case the expected number of bits is $\mathbb{H}_{ce}(p, q) = \mathbb{H}(p)$. This is also known as the Shannon's source coding theorem.

0.0.1 See also

0.0.2 References